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Establishing colour harmony evaluation and recommendation model for clothing colour matching based on machine learning and deep learning

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Abstract

Appropriate colour combinations improved aesthetic design quality and provided a comfortable and pleasant visual experience. However, applying current colour harmony models to clothing colour matching raised doubts about whether the existing theory needed to be refined, requiring further clarification with modern aesthetic perspectives. The study conducted a psychophysical experiment to investigate modern people's perceptions of colour harmony in clothing colour combinations. The results indicated that modern perceptions of colour harmony in clothing differed significantly from previous theories. For applications, two colour harmony models were established with unified fashion datasets for evaluating clothes matching based on colour harmony rules and observers' perceptions. The result showed that the rule-based model could accurately predict all items following nine colour harmony theories, and the perception-based colour harmony evaluation models aligned with contemporary aesthetic preferences were developed with a semi-supervised learning approach. The models, including support vector machines and custom convolutional neural networks, could predict colour harmony perception for the input images with high performance, and the model based on the generative adversarial network could provide colour recommendations for colour matching. The machine learning and deep learning model proposed in the study could be used for aesthetic judgment to generate clothing colour recommendations and provide valid design suggestions for clothing colour matching in the fashion and clothing industry.

Keywords: Colour harmony, Clothes matching, Colour combination, Colour design, Machine learning, Deep learning, Generative adversarial network

Introduction

In design aesthetics, colour harmony refers to colour combinations that evoke satisfaction, balance, and pleasure (Paterson, 2003). It aims to create a pleasing visual experience by blending two or more colours. Theories by Moon and Spencer define colour harmony as systematic variations producing pleasant coordination (Moon & Spencer, 1944), while Judd and Wyszecki describe it as the positive emotional response from

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adjacent or multiple colours (Judd et al., 1963). Colour combinations can elicit both positive emotions (excitement, relaxation) and negative ones (sadness) (Sutton & Altarriba, 2016). Thus, achieving harmonious colour schemes is a crucial goal for designers and researchers. Classic colour harmony methods, including those by Munsell, Moon, and Spencer, and the Natural Colour System (NCS) rely on specific colour systems. Munsell suggested using weak colours for large areas and strong colours for small ones. Moon and Spencer created aesthetic models to calculate indices for colour harmony based on order and complexity. The NCS focuses on consistency in attributes and brightness for harmonious effects. Colour harmony is also linked to eight characteristics: area, association, stylistic attitude, composition method, interaction, order, similarity, and hue (Burchett, 1991). However, these theories have limitations, prompting the development of simplified techniques for practical colour-matching efficiency, such as achromatic-chromatic parity and tonal, tone-on-tone, and tone-in-tone approaches.

Ou et al. (2004) studied emotional responses to 190 colour pairs across 11 dimensions, finding no significant cultural differences but notable gender differences. A colour emotion prediction model was developed based on colour appearance attributes, inferring emotions from combinations. Following this, the study conducted psychophysical experiments with 1,431 colour pairs to quantify colour harmony, asking participants to rate harmony on a scale of 1–10 in a dark environment. The resulting model effectively predicts colour harmony. The research also explored the definitions of colour harmony and its relationship to pleasure and order and compiled traditional theories in the 2015 Handbook of Colour Psychology. They identified common principles of colour matching, noting that combinations with the same hue or chroma tend to create harmony, slight brightness variations can reduce harmony, and higher lightness values correlate with greater harmony. However, the researchers emphasized the need for further study on how scenes affect colour harmony, highlighting the importance of practical applications in specific design contexts (Ou, 2015; Ou & Luo, 2006). Wicaksono and Ou conducted a psychophysical experiment with 20 observers to assess the colour harmony of 58 shirt and pants combinations. The findings showed a strong correlation between male and female assessments, and altering the colour combinations influenced observers' impressions and scores of harmony (Wicaksono et al., 2015). Liu and Ou explored colour harmony in clothing matching using images of Taiwanese youth, revealing significant effects of gender and style on psychological evaluations. The research noted that Ou's colour harmony model was less effective for dual-colour combinations, while Huang's model provided better predictions (Liu & Ou, 2017). Park examined the impact of social norms and colour associations in Korea on clothing choices and emotional responses, finding that wearing cheerful colours improved mood, and participants felt good in attractive colours (Park, 2017).

Weingerl and Javoršek's literature review emphasizes the complexity of selecting harmonious and effective colours in visual design, highlighting the importance of colour harmony in choosing, evaluating, and generating colour schemes (Weingerl & Javoršek, 2018). However, most research has focused on 2D colour configurations. Gong et al. explored colour harmony in 3D configurations to address real-life product design needs using 11 fundamental colours and five colour scheme techniques (Gong & Lee, 2019). Odabaşıoğlu and Olguntürk's study found that the area of indoor

spaces influences colour harmony for warm and cool colour combinations. However, the correlation between colour harmony and related terms was only moderately significant (Odabaşıoğlu & Olguntürk, 2020). Yaxyoxonovich analyzed clothing colours, recommending that gender, age, body type, and psychological type be considered in colour pairing (Yaxyoxonovich, 2021). Ng criticized the ambiguity of colour harmony as a teaching concept, suggesting that practical models like the Munsell colour system be used instead to teach colour differences more effectively (Ng, 2021).

Contemporary colour harmony models fall into two categories: those based on empirical rules by artists and designers and those constructed from collected samples to uncover latent features. These approaches have different perspectives and need a unified framework. To address this, Lu et al. developed a colour harmony model for image aesthetics using a Bayesian framework, integrating empirical rules as priors with features from machine learning (ML) methods to enhance aesthetic evaluation. They later optimized this model using a statistical learning framework, employing semantic labels and visual features in a Latent Dirichlet Allocation (LDA) framework to generate pleasing colour combinations (Lu et al., 2015, 2016a, 2016b). Rostami et al. assessed Iranian responses to monochromatic stimuli in the CIELAB colour space, finding that high lightness is linked to emotions like clean and soft. In contrast, high chroma relates to warm and modern feelings. They compared the reliability of colour emotion models with Ou's and Xin-Cheng's models, which excelled in different areas (Rostami et al., 2015). Yang used a simulated annealing optimization algorithm to generate colour combinations adhering to colour harmony principles, effectively applying it in mechanical tools (Yang & Zhang, 2015). Hsiao proposed a method to evaluate product image aesthetics using colour matching, achieving high satisfaction (Hsiao et al., 2017). Bai and Xue analyzed the relationship between colour schemes and emotions, identifying personality, gender, and fashion as key factors, with colour emotions influenced by hue warmth, saturation, and brightness (Bai & Xue, 2021).

Historically, colour data mining focused on individual colours rather than correlations between them. Goree and Crandall explored unsupervised generative modeling to analyze online colour scheme generators and real-world colour schemes (Goree & Crandall, 2020). Katayama et al. tested Nayatani's colour sensation prediction formula for kimono colour combinations, finding it effective but less accurate for contrast and pleasantness (Katayama et al., 2022). Wang et al. combined experimental psychology with AI to develop a mathematical model predicting colour harmony, finding that subjective factors had a more substantial influence than objective ones. Their research also identified significant correlations between various physical colour features and colour harmony, with linear models outperforming nonlinear ones in prediction accuracy and random forest models excelling in classification (Wang et al., 2022a, 2022b).

With artificial intelligence (AI) rapidly advancing, managing fashion big data has become a significant challenge. Wang and Rasheed's review highlights AI's growing role in fashion, initially grounded in basic fashion theories and leading to AI stylist models. The research examines neural networks, genetic algorithms, support vector machines, and other AI methods in fashion (Wang & Rasheed, 2014). Gu and Lou proposed an automatic webpage colouring framework that is validated in typical webpage design

scenarios, focusing on visual contrast, multi-colour compatibility, and semantic relevance (Gu & Lou, 2016).

Guan et al. reviewed 130 studies on clothing recommendations, identifying four main model types: clothing search/retrieval, wardrobe recommendation, fashion coordination, and intelligent recommendation systems. They noted gaps in research on clothing and user profiles (Guan et al., 2016). Hong et al. used AI tools for fashion design evaluation, particularly 3D-to-2D design for personalized fits (Hong et al., 2018). Lin et al. introduced the Neural Outfit Recommendation (NOR) framework, which improved recommendations and provided explanations through generated comments (Lin et al., 2019). Liu et al. used generative adversarial network (GAN) models to explore clothing matching based on semantic attributes, proposing Attribute-GAN for generating outfits (Liu et al., 2019). Gu et al. discussed AI challenges in fashion, including data standardization and evaluation (Gu et al., 2020). Shamoi et al. used the Fuzzy HSI model to predict aesthetic preferences and harmony, showing effective results in clothing colour harmony prediction (Shamoi et al., 2016, 2020). With the rise of online shopping, recommender systems have become crucial for personalized fashion suggestions. An et al. developed a fashion image dataset featuring hybrid styles and employed backbone models in machine learning to evaluate the accuracy of style recognition. The research's outcomes contributed to data-driven fashion services by enabling personalized recommendations for fashion style design (An et al., 2023). As current AI-based design generation solutions primarily focus on fashion trend analysis and the development of automated fashion image generation technologies, Choi et al. proposed an AI-based garment development system that incorporates domain knowledge from the fashion field. By comparing the design processes of human designers and AI systems, the approach aims to support fashion designers in garment development and design (Choi et al., 2023). Pan et al. proposed an optimized Swin Transformer model incorporating a Global Attention Mechanism, which enhances the model's ability to process global and local information in an integrated manner, which improved classification accuracy and robustness, addressing the challenges of automatic classification caused by the diverse styles of women's trousers and complex backgrounds (Pan et al., 2024). Suvarna and Balakrishna proposed a deep ensemble classifier combining five pre-trained models for classification accuracy to improve reliability. Using cosine similarity for recommendations, the model achieves 96% accuracy on benchmark datasets (Suvarna & Balakrishna, 2024).

The rise of deep learning (DL) has advanced research in fashion colour harmony, enabling more complex neural networks to extract latent features related to colour coordination. These models can improve prediction accuracy and suggest harmonious colour combinations through image generation. However, challenges still need to be addressed, such as the need for large standardized and labeled fashion datasets, the absence of models capable of simultaneous prediction of multiple color harmony rules, difficulties in learning effective feature representations, alignment with current colour harmony perceptions, and fashion evaluation complexities. The recent research on colour harmony related to two-piece clothing coordination dates back nearly a decade (Huang et al., 2017). Previous studies have indicated that colour harmony depends on factors influencing the relationship between colour and aesthetic response, such as individual and cultural differences, perception, context, and temporal characteristics (O'Connor,

2010). Hence, it is increasingly necessary to further explore contemporary perceptions of colour harmony and aesthetics in the current temporal and spatial context, and to evaluate the necessity of moving beyond rule-based colour harmony evaluation toward a new framework grounded in human perception. A psychophysical experiment will be conducted to investigate the subjective perception of different clothing colour combinations as a newly proposed perception-driven harmony evaluation, which can be a labeling method for existing fashion datasets, and used to refine and optimize the colour harmony model with these findings. By utilizing the aforementioned perception-driven labels, the image generative model produces recommended harmonious colour combinations in fashion contexts, offering meaningful contributions to developing human perception-based AI systems.

To address these issues and needs, developing fashion colour harmony evaluation and recommendation models aligned with contemporary harmony perception using deep learning on large-scale fashion datasets is essential. Therefore, the specific objectives of this study comprise the following four components, implemented through multiple empirical stages: 1. Develop a rule-based colour harmony evaluation model capable of simultaneously predicting various types of harmony in fashion colour combinations, addressing the absence of models that incorporate multiple harmony rules, and demonstrating the feasibility of training such colour harmony models. 2. Conduct a psychophysical experiment to examine the consistency between contemporary aesthetic preferences and existing rules of colour harmony. 3. Establish a perception-driven harmony evaluation model through semi-supervised learning, using experimental results as modern annotations for colour harmony. 4. Develop a generative model grounded in human perception for recommending harmonized fashion colour combinations. By leveraging the advanced development of AI technology, this research can provide more convenient and practical applications for the fashion design industry. The proposed colour harmony evaluation and recommendation model can effectively address the dilemmas and demands of everyday life regarding clothing colour-matching choices. Moreover, it can be a reference for the fashion industry in design processes.

Methods

Data preparation

This study employed the DeepFashion-Multimodal dataset (Jiang et al., 2022) to construct a colour harmony model. The dataset comprised 44,096 high-resolution human images, including 12,701 full-body samples with segmented masks. Given that the objective was to examine the harmony of colour combinations, images depicting individuals wearing one-piece garments were excluded. Consequently, 9,056 images were retained, ensuring a clear distinction between upper garments (e.g., shirts, blouses) and lower garments (e.g., pants, skirts, jeans). In the data preprocessing stage, the average RGB values and area proportions of the tops and bottoms were extracted from the segmented masks and converted to CIEXYZ and CIELCH colour spaces. For training the neural network, 20 features were selected as input variables, including RGB values, XYZ tristimulus values, CIE L^*, C^*, h^*, and the area proportions of both upper and lower garments. Based on established colour harmony principles, the model was designed to predict nine output labels comprising eight binary categorical variables and one continuous evaluation

metric. These principles included Munsell and Natural Colour Systems (NCS), achromatic-chromatic parity, tonal, tone on tone, tone in tone, and Moon and Spencer. Table 1 defines these colour harmony principles.

During the data preparation stage, the images in the dataset need to be annotated to develop a rule-based colour harmony evaluation model. Each image is labeled according to whether it conforms to each of the nine aforementioned colour harmony rules, resulting in eight binary classification labels and one numerical label per image. The specific approach involves using image processing techniques to extract each image's upper and lower garment regions, either by utilizing the mask annotations provided in the dataset or by applying the Segment Anything Model (Kirillov et al., 2023) for region segmentation. After segmentation, each region's area and average RGB values are calculated and converted into the CIELCH color space to obtain hue, lightness, and chroma features. These features can evaluate whether the image follows those colour harmony rules.

In the Munsell system, the area ratio between the upper and lower garments is inversely proportional to the product of their lightness and chroma. The region with a smaller area should have a higher lightness–chroma product, while the region with a larger area should have a lower product value, reflecting the principle that strong colours should be used in small areas and weak colours in large areas, in accordance with the harmony rule definition. The aesthetic score based on Moon and Spencer's theory is derived from the area of the extracted image regions, the number of colours, and the order differences in hue, lightness, and chroma. Determining each colour attribute difference interval with classification can retrieve corresponding scores from reference tables (Lu et al., 2015). These scores are then used in harmony rule formulas to compute the final aesthetic value. In the NCS, three types of consistency can contribute to a sense of harmony. For attribute consistency, two colours are considered consistent if their hue, lightness, and chroma values are the same or similar, with threshold values set at 30 for hue, 7.5 and 3.5 for lightness and chroma. For relational consistency, harmony is judged

Table 1 Definition of previous colour harmony rules

Theories	Definition and description
Munsell (Westland et al., 2007)	Large areas use weak colours, small areas use strong colours. $\frac{A_1}{A_2} = \frac{V_2C_2}{V_1C_1}$, where A represents the area of the two colours used, V represents the value of lightness, and C represents the value of chroma
Moon and Spencer (Lu et al., 2015)	Aesthetic model. $M = \frac{O}{C}$, where M represents the aesthetic index, O represents the order element, and C represents the complexity element
Natural Colour Systems (NCS) (Hård & Sivik, 2001; Park et al., 2010)	• Consistent Attribute: Colours with the same or similar colour attributes • Consistent Relationship: Colours with proportions of colour attributes that are the same • Consistent Lightness: Colours with the same brightness
Achromatic-chromatic parity (Weingerl & Javoršek, 2018)	Using achromatic colours as the primary colours, with chromatic colours as secondary accents
Tonal (Ou et al., 2018)	Adjacent colours or multiple colours are presented with similar hues and similar tones
Tone on tone (Munsell et al., 1969)	Adjacent colours or multiple colours are presented with the same hue but different tones
Tone in tone (Munsell et al., 1969)	Adjacent colours or multiple colours are presented with different hues but the same tones

based on proportional relationships among the three attributes—for example, if the ratio between the chroma and lightness or between the chroma and the sum of chroma and lightness for the two colours is less than a 0.01 difference. Lightness consistency requires the two colours to have the same lightness, with a threshold difference of no more than 5.

Under the achromatic–chromatic parity rule, two image regions' color combinations are evaluated to determine whether one is achromatic and the other chromatic. A colour is achromatic if its chroma value is less than 5. For the rules of tonal, tone on tone, and tone in tone, tone is defined as the product of lightness and chroma. In the tonal rule, two image regions are considered to have tonal harmony if they exhibit similar hue and tone, with similarity thresholds defined as a hue difference of 10–30 degrees and a tone difference of 5–25. The tone on tone and tone in tone harmony rules respectively examine whether the two colour regions have the same hue but different tones, and the same tone but different hues. A similar hue is defined as a hue difference of less than 10 degrees, while a similar tone is defined as a tone difference of less than 5. Figure 1 illustrates the distribution of samples adhering to each colour harmony rule, with respective counts of 4,315, 5,291, 537, 876, 1,329, 1,009, and 1,039. The establishment and results of the rule-based colour harmony evaluation model will be described in detail in the paper's Modeling section.

Psychophysical experiment

Experimental setup

A total of 20 observers, including 10 males and 10 females, participated in this experiment. The mean age of the observers was 20 years, with a standard deviation of 0.94. All participants were required to provide informed consent before participation. The study was approved by an ethics committee or institutional review board, with the approval number NCKU HREC-E-113-154-2. Before the experiment, all 20 observers

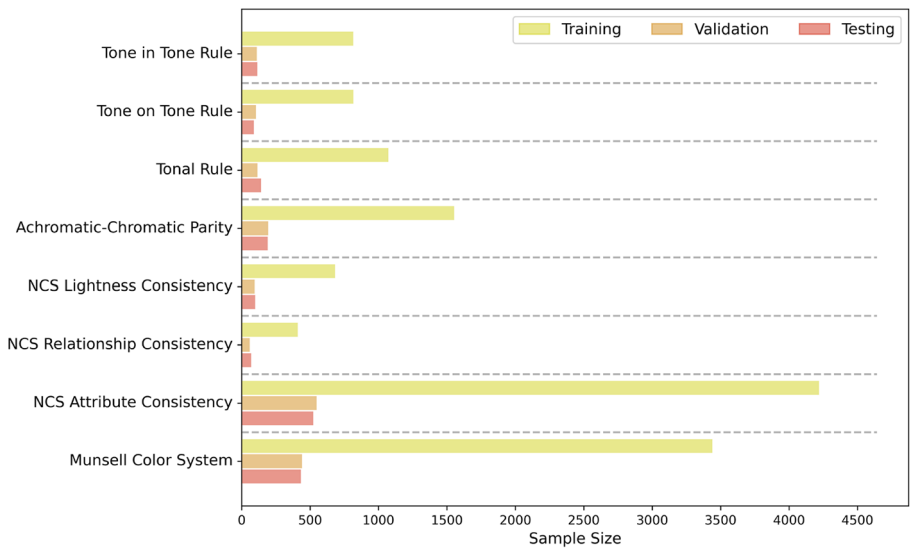


Fig. 1 Number of samples following different colour harmony rules

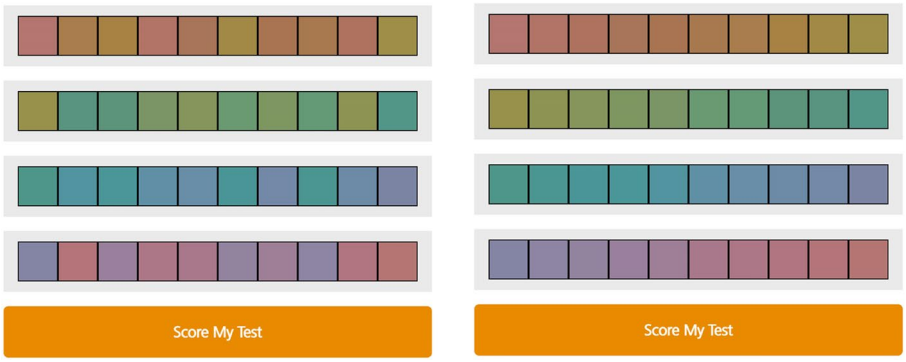
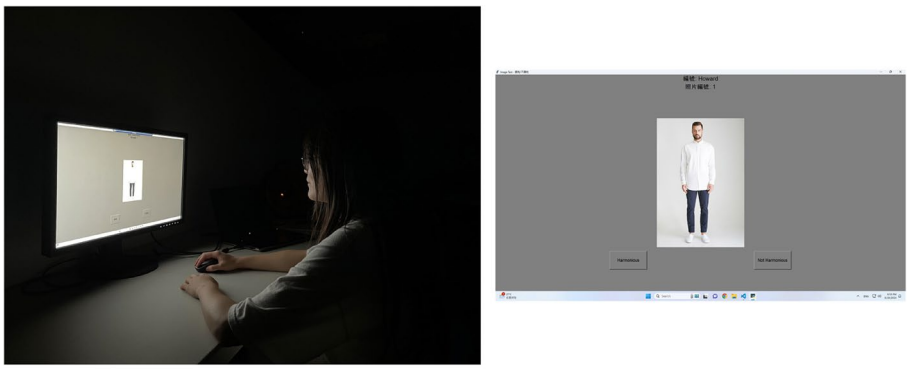


Fig. 2 The X-rite colour challenge and hue test



(a) (b)

Fig. 3 **a** Experimental condition; **b** A screenshot of experimental UI

completed the X-Rite Colour Challenge and Hue Test (Murgård & Ståhl, 2023) using the same display as in the experimental setup. Figure 2 presents the test interface. The initial arrangement of colour blocks was displayed on the left side, with the first and last blocks fixed in position. The participant was required to reorder the colour blocks in each row based on hue variation. This process was repeated until all four rows were arranged. The right side of the interface displayed the final arrangement after completion. The observers achieved an average test score of 2; lower scores indicated a lesser degree of colour perception impairment.

The experiment employed a 27-inch professional colour-calibrated monitor (EIZO ColourEdge CS2731), which supported 99% of the Adobe RGB colour gamut. This display was widely utilized in various fields requiring high colour accuracy. Before the experiment, the display was calibrated using the X-Rite i1Basic Pro 3 spectrophotometer to ensure precise colour reproduction of the test images. Following calibration, the display achieved a maximum luminance of 120 cd/m^2 with a reference white point set to the D65 illuminant, corresponding to CIE1931 (x, y) chromaticity coordinates of (0.3127, 0.3290). The experiment utilized 500 full-body images selected from the DeepFashion-Multimodal dataset as test stimuli. Each test image contained upper garments (e.g., shirts, blouses) and lower garments (e.g., pants, skirts, jeans).

Figure 3a presents the experimental setup. A user interface (UI) was implemented to facilitate data collection, as shown in Fig. 3b, to record observer responses automatically. The test image was displayed at the center of the interface, while all surrounding areas were presented in neutral gray to minimize potential interference from simultaneous contrast effects. Figure 4 depicts the CIE1931 chromaticity coordinates of the tops and bottoms in the test images.

Experimental procedures

The experiment took place in a completely dark environment. Before the experiment, each observer underwent a 20-min dark adaptation period. The viewing distance between the observers and the display was 60 cm. During the experiment, observers evaluated whether the clothing combinations on the monitor exhibited colour harmony. They indicated their judgment by selecting the corresponding button for harmonious or non-harmonious colour combinations. The response was recorded once a selection was made, and the next test image appeared. This process continued until all 500 images were evaluated. The test images were presented in a randomized order to enhance result reliability. Each observer completed the experiment twice to assess the consistency of the responses. An interval between the two sessions was 1 week to ensure the reliability of the experimental data.

Results and Discussion

Observer variability

To ensure the accuracy of experimental data and provide reliable analytical results, this study employs two indicators—Hamming distance and Jaccard coefficient—to evaluate inter-observer and intra-observer variability. Inter-observer variability measures the consistency between an individual observer's assessments and the average assessment of all observers. In contrast, intra-observer variability quantifies the similarity between an observer's first and second assessments. Equations (1) and (2) define the calculations for Hamming distance and the Jaccard coefficient, respectively. Hamming distance is the

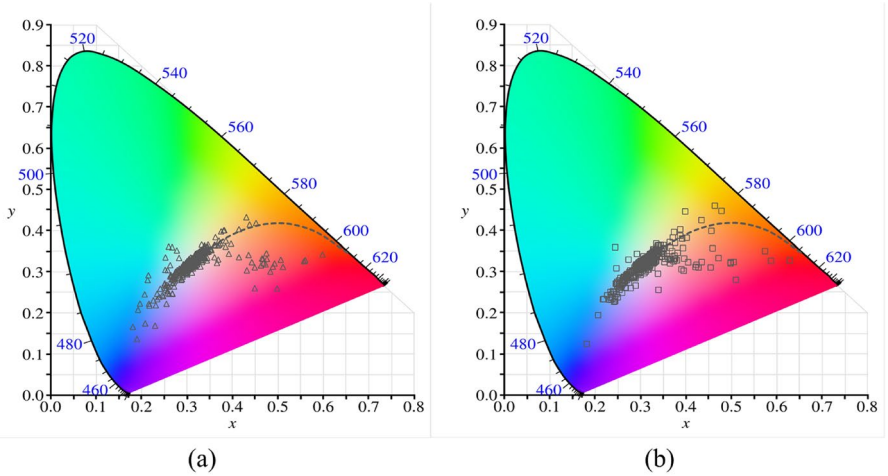


Fig. 4 CIE1931 chromaticity coordinates of test images **a** Tops; **b** Bottoms

number of positions at which two datasets differ, where x and y represent different datasets with an equal number of data entries, n. It can be computed using an XOR operation between x and y and serves as a metric for comparing two binary datasets. A higher Hamming distance indicates lower consistency between the datasets, whereas a lower Hamming distance suggests greater agreement, which is preferable for consistency. Conversely, the Jaccard coefficient evaluates the similarity between two sets, denoted as S_A and S_B. It is defined as the ratio of the intersection to the union of the two sets. A higher Jaccard coefficient signifies a greater degree of similarity between the datasets.

$$D_H(x, y) = \sum_{i=1}^{n} |x_i - y_i|, \; if \; x = y \Rightarrow D = 0; \; if \; x \neq y \Rightarrow D = 1 \tag{1}$$

$$C_{\text{jaccard}}(S_A, S_B) = \frac{|S_A \cap S_B|}{|S_A \cup S_B|} \tag{2}$$

Inter-observer variation quantifies the consistency between an individual observer's and average ratings across all observers. The average rating is determined using majority voting for each instance. In contrast, intra-observer variation measures the average difference between an observer's first and second ratings across all instances. Table 2 presents the average values of inter-observer and intra-observer variation. The analysis of these variations indicates that both fall within acceptable ranges, confirming the reliability of the experimental data.

Comparison to colour harmony theories

The study compares experimental data with results from various representative harmony rules to examine whether contemporary perceptions of colour harmony in fashion align with established colour harmony principles. In the analysis, experimental data obtained from observers must first be filtered to exclude instances with excessive inter-observer and intra-observer variation. Subsequently, majority voting and average ratings are computed to enhance the accuracy of the subsequent analyses.

To assess the consistency of colour harmony classifications, the Hamming distance and the Jaccard coefficient are employed to evaluate the similarity of binary classification-based harmony rules. Since the Moon and Spencer model determines colour harmony using aesthetic scores as continuous values, the Pearson correlation coefficient is calculated between the average experimental ratings and Moon and Spencer's scores to quantify their relationship.

The consistency between the experimental results and various colour harmony rules is presented in Table 3. The findings indicate that contemporary perceptions

Table 2 Definition of previous colour harmony rules

	Hamming distance	Jaccard coefficient
Inter-observer	73	0.852
Intra-observer	57	0.884

Table 3 Definition of previous colour harmony rules

Theories	Hamming distance	Jaccard coefficient
Munsell	241	0.518
NCS-Attribute	168	0.664
NCS-Relationship	465	0.070
NCS-Lightness	410	0.180
Achromatic-chromatic parity	379	0.242
Tonal	358	0.284
Tone on tone	424	0.152
Tone in tone	408	0.184

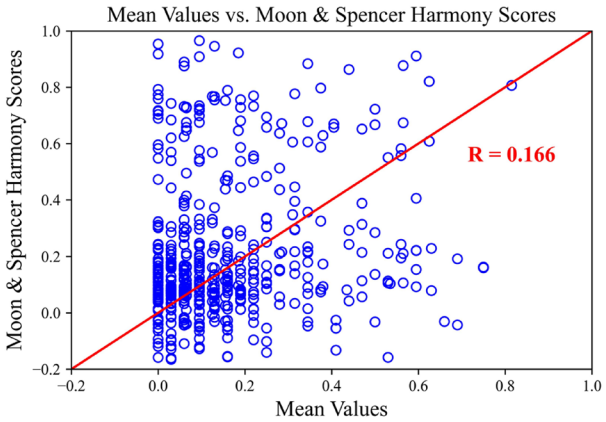


Fig. 5 Correlation between the mean value of observers' response and Moon and Spencer rule

of clothing colour harmony deviate from traditional theories, particularly in the case of the Natural Colour System (NCS) and tone-on-tone harmony, which exhibit relatively higher and lower consistency, as reflected by their respective Hamming distances and Jaccard coefficients. The NCS model demonstrates the highest consistency with observer ratings among these rules. However, the Pearson correlation coefficient between the inverse of the mean subjective evaluations and the Moon and Spencer model is only 0.166, suggesting a considerable discrepancy between this rule and modern perceptions of clothing colour harmony. Figure 5 presents a scatter plot comparing the visual averages with predictions based on the Moon and Spencer rule. Figure 6 illustrates colour pairs from the experiment identified as harmonious or non-harmonious, mapped in the CIELAB colour space. While some pairs still partially conform to specific colour harmony principles, the results suggest that existing models do not fully capture contemporary aesthetic preferences. These findings highlight the need to develop an evaluation model for clothing colour harmony that better aligns with modern aesthetic standards.

The study also compares the most recent colour harmony models, including those by Huang (2017) and Ou (2018). Since the colour harmony perception values in this study range from 0 to 1, correlation coefficients are adapted to compare the predictive performance of the two colour harmony models. The experimental results show

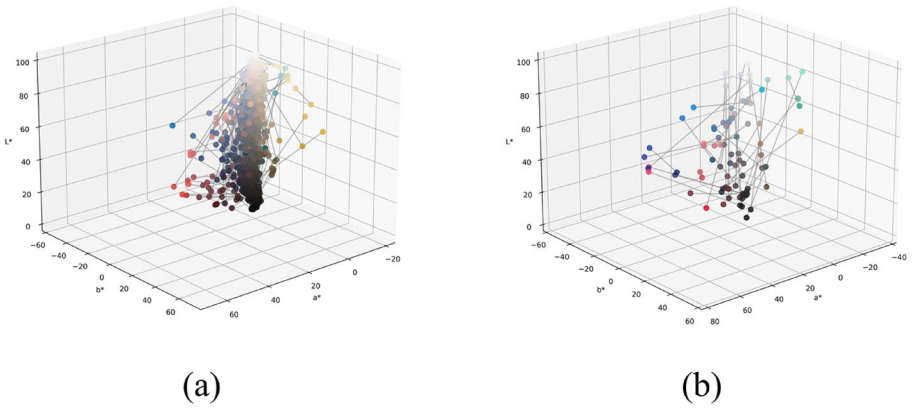


Fig. 6 Colour pair of **a** colour harmony; **b** not colour harmony in the experiment

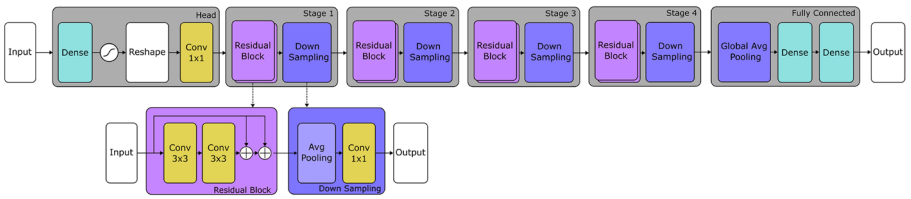


Fig. 7 Architecture of the deep neural network

that the correlation between Huang's model and the observers' perception in the experiment is 0.11, while the correlation with Ou's model is −0.02. It indicates that the colour harmony definitions in these two models have diverged from current perceptions, with Ou's model being almost non-correlated. The primary reason might be that Ou's model is not designed explicitly for clothing colour combinations and does not account for the impact of different area proportions. On the contrary, since Huang's model is derived based on color combinations of two-piece garments, its slightly higher relevance to this study's results than Ou's model may be attributed to both focusing on the harmony of clothing color combinations. However, the overall lower correlation of Huang's model to our study may be due to its consideration of two different age groups, including those under 45 and those over 45, and its experimental results indicated that these age groups perceive color harmony differently. As our study primarily involved participants around 20 years old, this age difference may have led to noticeable discrepancies in model prediction. Therefore, it underscores the necessity of developing a colour harmony model based on modern aesthetics.

Modeling

Rule-based colour harmony evaluation model

The study uses inputs from 20 features obtained during the data preparation to develop a rule-based deep learning model. The outputs are the results of assessments based on nine colour harmony rules, including eight binary classifications and one continuous value. The created model can simultaneously output all nine items. The dataset is split into training, validation, and test data in an 8:1:1 ratio.

As illustrated in Fig. 7, the network architecture consists of a head and four preceding stages. The head comprises a fully connected (FC) layer with $4 \times 16 \times 16$ neurons, followed by a sigmoid activation function, and then reshaped into a [N, 4, 16, 16] tensor. A convolutional layer with a kernel size of 1, stride 1, and zero padding is applied to expand the four channels into 80 channels without any learnable bias. Each preceding stage consists of two residual blocks containing two mini-convolution blocks and a down-sampling block. Consequently, each stage incorporates four convolutional layers, similar to a standard ResNet block, with two convolutional layers having a kernel size of 3, stride 1, and padding of 1. The output of each block is summed with itself, passed through a GELU activation function, and then added to its value before being fed into the block. The down-sampling block consists of average pooling with a size of 2 and stride 2, followed by a no-bias convolutional layer with kernel size 1, stride 1, and zero padding. The network utilizes four stages, with the number of channels increased by an expansion factor (k) of 2. The final output is globally average pooled and flattened before being passing through an FC layer with 320 neurons, followed by a Mish activation function. The output is then reduced to 9 classes for loss calculation via a final FC layer. The hyperparameters for batch size, learning rate, and optimizer are set to 256, 10^{-3}, and AdamW with AMSGrad enabled for training, with early stopping employed as a regularization technique.

In this study, several metrics are employed to assess the model's performance in both classification and regression tasks, including macro accuracy, precision, recall, specificity, F1 score, as well as root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2). Furthermore, the receiver operating characteristic (ROC) curve demonstrates the classification model's performance across all categories, as shown in Fig. 8a. The experimental results indicate that the deep learning model can accurately predict the eight categories of colour harmony rules and one regression task. The average classification metrics for the eight colour harmony rules are as follows: macro accuracy 0.98, precision 0.96, recall 0.96, specificity 0.98, and F1 score 0.96, as presented in Table 4. The RMSE, MAE, and R^2 for the regression results are 0.06, 0.03, and 0.92, respectively. Figure 8b displays scatter plots illustrating the predicted performance of the Moon and Spencer aesthetic index. Based on the above results, this study first develops a model capable of accurately predicting multiple color harmony rules simultaneously, demonstrating that color

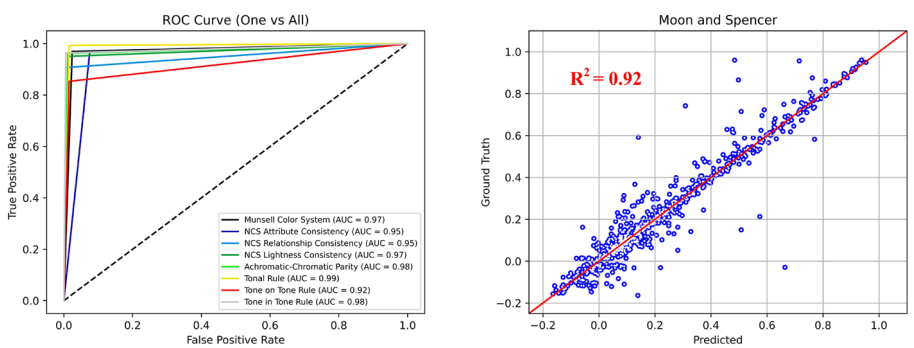


Fig. 8 Model performance **a** the ROC curves; **b** scatter plots of prediction

Table 4 Evaluation metrics of model performance for different colour harmony rules

Rules	Accuracy	Precision	Recall	Specificity	F1 score
Munsell	0.97	0.97	0.97	0.97	0.97
NCS-attribute	0.95	0.93	0.96	0.93	0.94
NCS-relationship	0.95	1.00	0.90	1.00	0.95
NCS-lightness	0.97	0.99	0.94	0.99	0.97
Achromatic-chromatic parity	0.97	0.99	0.95	0.99	0.97
Tonal	0.99	1.00	0.97	1.00	0.99
Tone on tone	0.93	0.98	0.89	0.98	0.93
Tone in tone	0.98	1.00	0.96	1.00	0.98
Mean	0.964	0.983	0.943	0.983	0.963

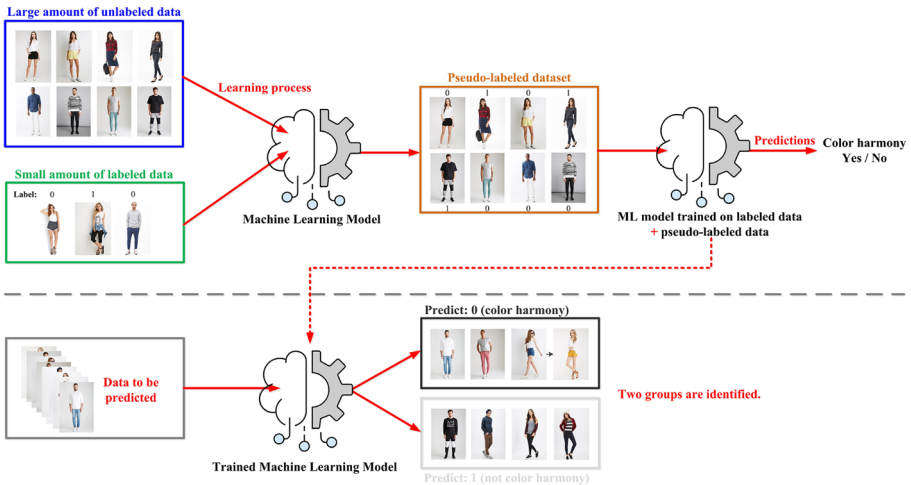


Fig. 9 Procedure of semi-supervised learning approach

harmony rules can be effectively modeled using machine learning and deep learning approaches.

Perception-based colour harmony evaluation model

Based on the experimental data, the study establishes a model to evaluate colour harmony in clothing matching. A semi-supervised learning (SSL) approach, aiming to develop a more robust evaluation model, is employed to train the model using only 500 labeled data points from the experiment. The implementation process of SSL is illustrated in Fig. 9. Initially, a small set of labeled data is utilized to train the model. After training, the model is used to predict unlabeled data. If the predictions fall within a specified threshold range, these predictions are treated as pseudo-labels for the unlabeled data. Subsequently, the model is retrained using both the small labeled dataset and the larger pseudo-labeled dataset, thus completing the SSL procedure.

Machine learning model

The initial step in building a machine learning model based on supervised learning is determining the model's input and output. The study used 20 features obtained during

Table 5 Performance of machine learning models

Models	Accuracy	Precision	Recall	F1 score
LR	0.78	0.80	0.78	0.78
KNN	0.84	0.88	0.85	0.84
GNB	0.65	0.66	0.65	0.65
SVM	0.98	0.98	0.98	0.98
DT	0.83	0.84	0.84	0.83
RF	0.94	0.95	0.94	0.94
ANN	0.95	0.95	0.95	0.95
XGboost	0.94	0.95	0.94	0.94

Table 6 Evaluation metrics of model performance

Metrics	Supervised	Semi-supervised
Accuracy	0.98	0.99
Precision	0.98	0.99
Recall	0.98	0.99
F1 score	0.98	0.99

the data preparation phase as inputs for the model. These features include average RGB values, their conversions to CIEXYZ and CIELCH colour spaces, and the percentages of the top and bottom areas. The output is a binary classification based on evaluations by observers from the experiment, where a value of 0 indicates colour harmony and a value of 1 indicates colour disharmony. The model is built using 500 data points from psychophysical experiments. The data distribution in these experiments reveals a notable imbalance between the labeled categories of harmonious and non-harmonious samples. To address this issue, the SMOTE-ENN algorithm, which integrates SMOTE (Synthetic Minority Oversampling Technique) and ENN (Edited Nearest Neighbors), is applied. This oversampling technique generates synthetic samples that better represent the minority class and reduces the presence of noisy samples, thereby enhancing the model's generalization ability. The input features undergo preprocessing through z-score normalization. The dataset is divided into training and testing sets with a 0.8 to 0.2 split. To ensure a fair evaluation of the model's performance, a stratified shuffle split cross-validation approach is employed, maintaining the class distribution in each split to match that of the original dataset.

The study compares various representative machine learning models., including Logistic Regression (LR), K-Nearest Neighbor (KNN), Gaussian Naive Bayes (GNB), Support Vector Machines (SVM), Decision Trees (DT), Random Forest (RF), Artificial Neural Networks (ANN), and eXtreme Gradient Boosting (XGboost). Grid search is applied to determine the optimal hyperparameter combination for each model. Table 5 shows the performance of various machine learning models. Experimental results reveal that the SVM model achieves the best performance with hyperparameters C set to 10 and gamma set to 1. The results in the supervised column of Table 6 indicate that all metrics, including accuracy, precision, recall, and F1 score, exceed

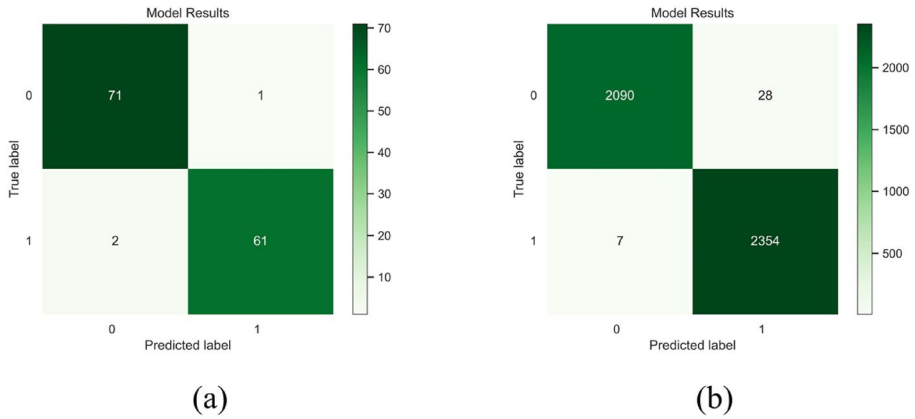


Fig. 10 Confusion matrix of the models **a** supervised; **b** semi-supervised

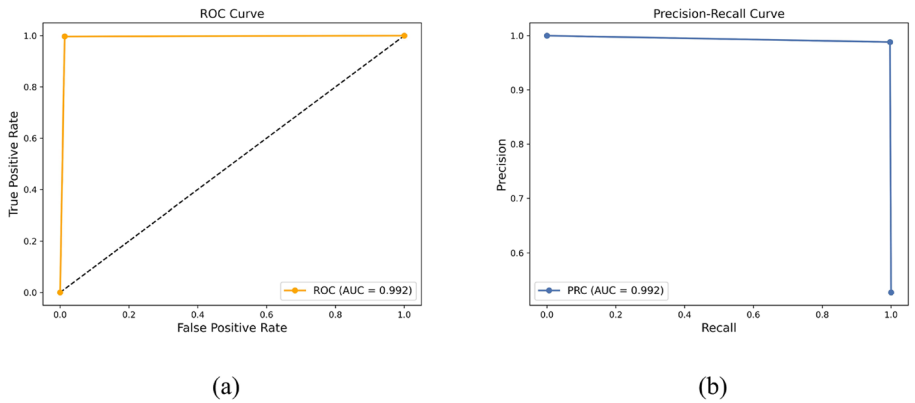


Fig. 11 ROC and PR curve of the semi-supervised model

0.98. As a result, this model is used to predict the pseudo-labels for the unlabeled data.

In the semi-supervised learning process, the labeled and pseudo-labeled data are merged to form a dataset of 9,056 entries. However, this dataset still exhibits class imbalance, prompting the use of the SMOTE-ENN algorithm for sampling. The dataset is divided into training and testing sets with a 0.7–0.3 split. Cross-validation is conducted by partitioning the data into ten folds for training. The results reveal that the SVM model, with hyperparameters C and gamma set to 1, achieves the best performance, with all evaluation metrics reaching 0.99. The SSL training method effectively predicts the subjective harmony perception of clothing colours. Figures 10 and 11 present the confusion matrices for supervised and semi-supervised models and the ROC and PRC curves. The semi-supervised learning approach combines supervised and unsupervised learning and provides an accurate and reliable model for evaluating clothing colour combinations, aligning with contemporary aesthetic standards.

Deep learning model

The initial step for training a deep learning model is to determine the model's input and output. The input consists of 9,056 images selected from the DeepFashion-Multimodal

dataset, and the output is a binary classification result indicating whether the colour combination is harmonious. This study finds that the ratio of harmonious to non-harmonious images in the dataset is approximately 9:1, indicating an imbalance that can bias the model towards the majority class during training, affecting evaluation accuracy. Hence, the SMOTE-ENN algorithm is also applied to balance the data. The balanced dataset is then split into training and testing sets in an 8:2 ratio, and stratified k-fold cross-validation is used on the testing set to monitor for overfitting during training, where k is 10.

Regarding model architecture, the study tested a custom convolutional neural network (CNN) architecture and two pre-trained models, including VGG16 and ResNet101, with transfer learning to evaluate their performance. A custom CNN model contains three convolutional layers with 32, 64, and 128 feature maps. A 2×2 max pooling layer follows each convolutional layer. After processing through the third max pooling layer, the image is flattened into a fully connected layer with a vector length of 3,584, followed by a hidden layer with 256 neurons and an output layer with a single neuron. The optimizer used is Adam, and binary cross-entropy is employed as the loss function for model training. The performance of the classification model is discussed using evaluation metrics such as accuracy, precision, recall, and F1-score, where the values are 0.953 for custom CNN. Besides, the confusion matrix, ROC, PRC, and the area under the curve (AUC) are also adopted, as shown in Fig. 12a–c, respectively. As a result, all the AUCs approach about 0.95, indicating that the model can accurately classify the input image as colour harmony (0) or non-harmony (1). The custom CNN performs better than the other two pre-trained models, likely because the appearance and colour of clothing are not

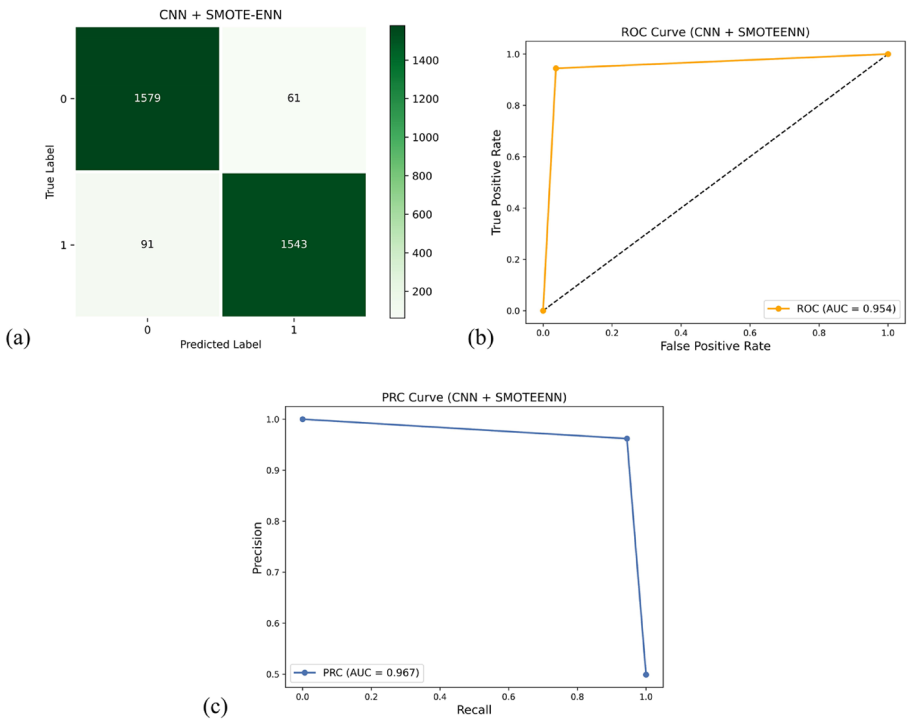


Fig. 12 Model performance **a** confusion matrix **b** ROC **c** PRC

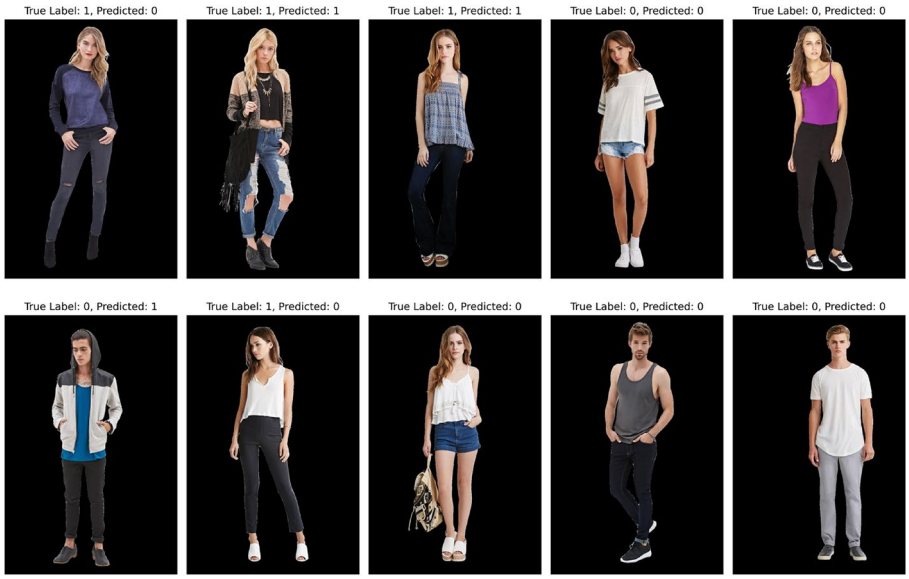


Fig. 13 Prediction results of the custom CNN model

high-level features. As a result, it does not require reliance on overly complex convolutional structures. Figure 13 shows the results of model prediction.

Comparison of ML and DL models

The study uses subjective evaluations obtained from experiments as the target for model prediction and employs a semi-supervised learning approach to develop a perception-based colour harmony assessment model. The model is created using feature extraction combined with a machine learning classifier to build the prediction model and a CNN to extract features automatically. Based on the model performance in experiments, the feature extraction combined with a classifier model achieved a higher prediction accuracy of 0.99, while the CNN-based approach attained an accuracy 0.95. Although both models demonstrated ideal performance, this study suggests that the CNN-based model is more convenient for practical applications, as users only need to input an image. The model can determine whether the cloth's colour combination adheres to colour harmony. On the other hand, if the feature extraction-based approach is adopted, additional work is required to develop an accurate clothing region detection model to extract colour feature areas precisely. Therefore, this study concludes that the perception-based CNN deep learning model offers greater application value and efficiency.

Recommendation model for clothing colour matching

The study utilizes a GAN to develop a recommendation model for clothing colour coordination. The recommendations are based on harmony labels obtained from experiments conducted in this research. The model can generate suggested clothing colour combinations from grayscale outfit images, ensuring that the results align with modern principles of colour harmony perception.

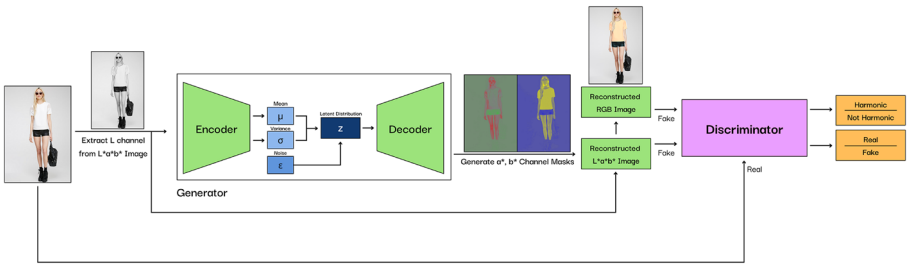


Fig. 14 Model architecture for recommendation of clothing colour matching

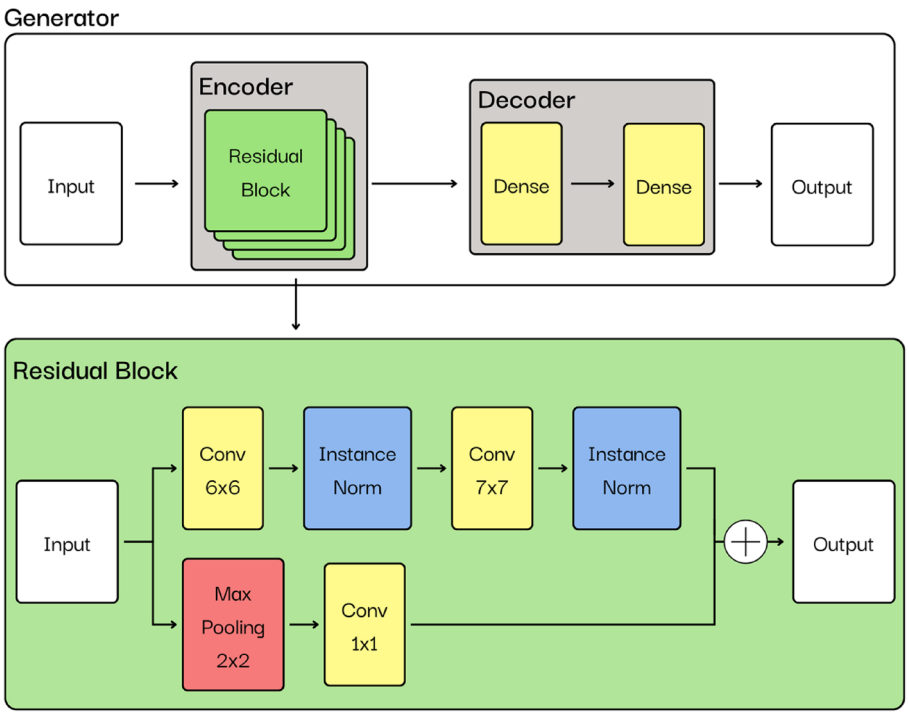


Fig. 15 Architecture of the generator

The data preprocessing pipeline includes segmenting tops and bottoms and extracting RGB and LAB colour information from segmentation masks. The model architecture consists of a generator and a discriminator, as shown in Fig. 14, both designed with a base number of 32 channels and a latent size of 256 and trained using the DeepFashion-Multimodal dataset with a fixed random state of 43 to ensure reproducibility. The dataset, containing 9,056 samples, is split into training, validation, and testing sets in an 8:1:1 ratio. An early stopping is applied to improve training stability and monitor the generator loss with a patience of five epochs. The training process is optimized using a batch size of 32, a learning rate of 10^{-5} for the generator, 5×10^{-5} for the discriminator, and a weight decay of 10^{-2}. The generator and discriminator utilize the AdamW optimizer with AMSGrad enabled, with beta1 set to 0.5. The loss function consists of binary cross-entropy applied to both the generator and discriminator, along with Kullback–Leibler (KL) divergence for the generator. The total loss is computed as the sum of binary

cross-entropy with logits loss and KL divergence loss, ensuring stable and meaningful latent space learning.

The generator is implemented as a Variational Autoencoder (VAE) comprising an encoder and a decoder. Figure 15 illustrates the architecture of the generator. The input to the generator is the L^* channel of an LAB image, which is processed through an encoder composed of four residual blocks. Each residual block includes a 6×6 convolution followed by instance normalization and a SiLU activation function, along with a 7×7 convolution and another instance normalization in the main branch. A transition branch performs a 2×2 max pooling followed by a 1×1 convolution. The outputs from both branches are summed and passed through another SiLU activation function. After passing through these four residual blocks, the encoder predicts the mean (μ) and standard deviation (σ) and fed alongside noise (ε) sampled from a normal distribution (mean = 0, variance = 1) into the decoder.

The decoder reconstructs the image from this latent representation using two dense layers, each with neurons matching the latent size. The final dense layer outputs a single ab value, using a Tanh activation function to align with the $[-128, 128]$ range of the LAB colour space. The predicted a^* and b^* channels are combined with the input L channel to reconstruct a full LAB image while preserving the texture of the original clothing. This generated LAB image is then converted to RGB to facilitate further evaluation by the discriminator. The generator ultimately produces four images per clothing item: the top and bottom in both RGB and LAB colour spaces.

The discriminator evaluates these generated images to determine their realism and harmony. It processes the LAB and RGB clothing pairs through four convolutional blocks, each consisting of a 4×4 convolution, instance normalization, and a SiLU activation function, as shown in Fig. 16. After the convolutional layers, global max pooling and average pooling are applied to extract relevant feature information. These pooled features are then passed through a bilinear layer, which simultaneously fuses the extracted features and predicts whether the clothing pairs are real or fake and whether they exhibit harmony. This design ensures that the discriminator effectively distinguishes between realistic and artificial clothing pairs while learning a meaningful representation of visual harmony. Figure 17 shows the results of the clothing colour matching recommendation. The top row presents the input grayscale images, while the bottom row displays the GAN-generated colourization results.

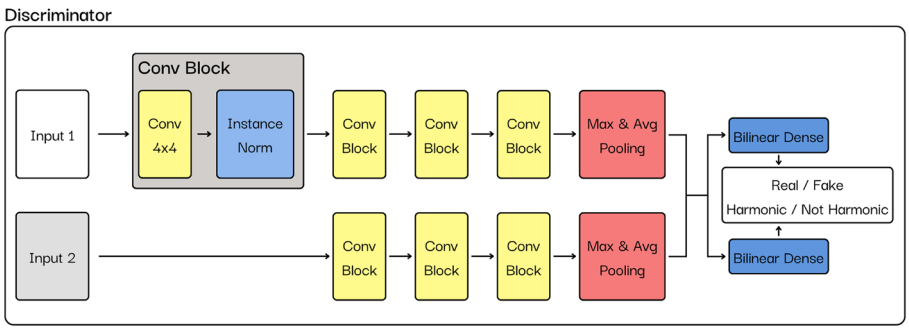


Fig. 16 Architecture of the discriminator



Fig. 17 Clothing colour matching recommendation (Top: grayscale images; Bottom: generated colourization results)

Conclusions

In conclusion, this study aims to develop colour harmony evaluation and recommendation models for clothing coordination using a comprehensive set of unified fashion datasets, examine subjective colour harmony perception across various colour combinations through a psychophysical experiment, and propose an optimized model. The experimental results demonstrate that contemporary colour harmony perception differs notably from traditional colour harmony theories, especially regarding the three rules: consistent relationships in NCS, tone-on-tone, and the Moon and Spencer aesthetic index.

To provide a more efficient colour harmony evaluation of an image, the study proposes a rule-based deep learning model based on a large, standardized fashion dataset that can be used to evaluate whether clothing colour combinations align with nine representative colour harmony theories first. Using a CNN architecture to extract critical features. The model can predict colour harmony categories and corresponding values with different clothing colour combinations. According to the experimental results, the established model can accurately predict whether the clothing combinations conform to colour harmony and can also be utilized to predict aesthetic indices, where the average values of the classification evaluation metrics are all above 0.94, and the coefficient of determination for the regression reaches 0.92. A semi-supervised learning approach is applied to develop perception-based colour harmony models that adhere to contemporary aesthetic standards using the data obtained from the experiment as labeled data. The resulting SVM machine learning model and custom CNN model can accurately assess whether clothing combinations demonstrate colour harmony, achieving high accuracy, precision, recall, and F1 scores, with classification evaluation metrics exceeding 0.99 and 0.95, respectively. In addition, the study adopts a GAN-based framework to develop a recommendation model for clothing colour matching, capable of generating colour combinations that align with current colour harmony perception.

However, this study also has some limitations. Due to the relatively small number of participants and the lack of consideration for the impact of racial and cultural differences, the generalizability of the findings is limited. Additionally, the study only explored color harmony between two primary clothing color combinations. In today's environment of diverse fashion colors, more complex color combinations need to be considered.

Future research could be strengthened in the following areas: 1. Increase the diversity of participants, such as different age groups and ethnicities, to provide more generalizable prediction and recommendation models, and further assess the influence of cultural differences on color harmony perception. 2. Consider more complex clothing color combinations, including increasing the number of colors in the combination, lighting conditions, types of garments, and styles, to develop broader and more reliable models. 3. Integrate the model with real-world visual environments and validate its effectiveness among culturally diverse user groups and in practical fashion applications.

Despite these limitations, the results of this study still provide valuable insights for the fashion industry, particularly in clothing design, by offering guidance on colour coordination and aesthetic considerations. The model can be utilized for aesthetic evaluation to generate clothing colour recommendations based on colour harmony principles, offering practical suggestions for colour matching. The experimental results, derived models, and recommendation system can provide the ideal colour design for fashion industry applications and be a reference for international organizations related to colour to formulate standards.

Abbreviations

ML	Machine learning
LDA	Latent dirichlet allocation
AI	Artificial intelligence
GAN	Generative adversarial network
DL	Deep learning
NCS	Natural colour system
UI	User interface
FC	Fully connected
RMSE	Root mean square error
MAE	Mean absolute error
ROC	Receiver operating characteristic
SSL	Semi-supervised learning
SMOTE	Synthetic minority oversampling technique
ENN	Edited nearest neighbors
LR	Logistic regression
KNN	K-Nearest neighbor
GNB	Gaussian Naive Bayes
SVM	Support vector machines
DT	Decision trees
RF	Random forest
ANN	Artificial neural networks
XGboost	eXtreme gradient boosting
CNN	Convolutional neural network
AUC	Area under the curve
KL	Kullback–Leibler
VAE	Variational autoencoder

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Some of the study's results have been published at ISPACS 2024 and ICCE-TW 2024. The article is an extended version, providing a more comprehensive narrative, including enhanced rigor in experimental methods, analysis, results, and modeling.

Author contributions

HCL* conceptualized the study, developed the methodology, validated the results, conducted the formal analysis and investigation, curated the data, administered the project, acquired funding, prepared the original draft, and reviewed and edited the manuscript. LKW, YKC, and KYH wrote the software, conducted the experiment, built the models.

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Data availability

The datasets used and/or analysed during the current study and the derived models are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study was approved by the National Cheng Kung University Human Research Ethics Committee (HREC) (NCKU HREC-E-113-154-2), with an approval date of 30 April 2024.

Competing interests
The authors declare that they have no competing interests.

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